BIXOLON®

Software Manual

Virtual COM for USB Driver / Configuration Tool

Rev. 1.23

SRP-S300
SRP-380 / SRP-382
SRP-F310II / SRP-F312II / SRP-F313II
SRP-350plusIII / SRP-352plusIII
SRP-350III / SRP-352III
SRP-330 / SRP-330II / SRP-332II / SRP-340II / SRP-342II
SRP-350II / SRP-350IIK / SRP-350plusII / SRP-352plusII
SRP-350 / SRP-350plus / SRP-352plus
SRP-F310 / SRP-F312
STP-103II / STP-103III
SRP-270 / SRP-275 / SRP-275II / SRP-275III
SRP-280 / SRP-340 / SRP-370 / SRP-372 / SRP-500

SLP-TX400 / SLP-TX403 / SLP-TX420 / SLP-TX423
SLP-TX220 / SLP-TX223
SLP-DX420 / SLP-DX423 / SLP-DX220 / SLP-DX223
SLP-D420 / SLP-D423 / SLP-D220 / SLP-D223
SLP-DL410 / SLP-DL413
SRP-770II

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1. Manual Information

This manual provides information on the usage of the “Virtual COM for USB Driver” and “Configuration Tool” software associated with USB interface use on BIXOLON printers.

This software can be used to set and use the USB interface via the serial communication mode.

※ The Latest file versions is available for download from the BIXOLON website (www.bixolon.com).

We at BIXOLON maintain ongoing efforts to enhance and upgrade the functions and quality of all our products. In following, product specifications and/or user manual content may be changed without prior notice.

2. Usage Environment

2-1 Operating System Environment

Microsoft Windows XP SP3 (32bit, 64bit)
Microsoft Windows Server 2003 SP1 (32bit)
Microsoft Windows Server 2003 (64bit)
Microsoft Windows VISTA (32bit, 64bit)
Microsoft Windows Server 2008 (32bit, 64bit)
Microsoft Windows Server 2008 R2 (64bit)
Microsoft Windows 7 (32bit, 64bit)
Microsoft Windows 8 (32bit, 64bit)
Microsoft Windows Server 2012 (64bit)
Microsoft Windows 10 (32bit, 64bit)

2-2 Supported Printer Models

The “Virtual COM for USB Driver” software can be used only with USB-supported printers.

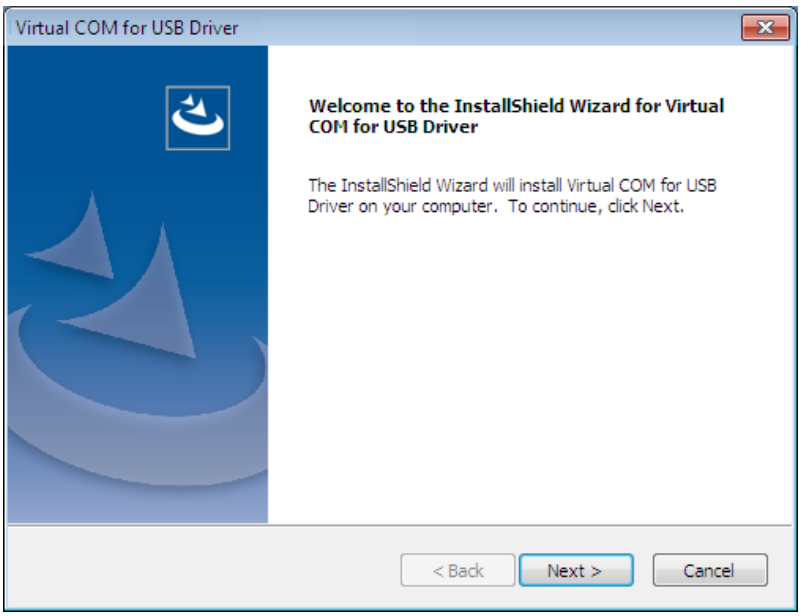
3. Software Installation & Uninstallation

*** Installation and uninstallation process in this document is written based on Windows 7**

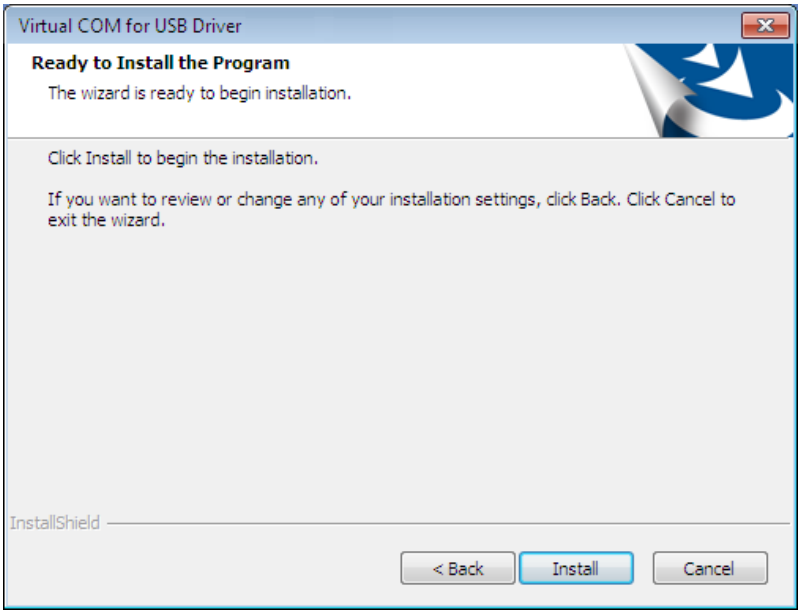
3-1 Installation

1) Execute the installation file.

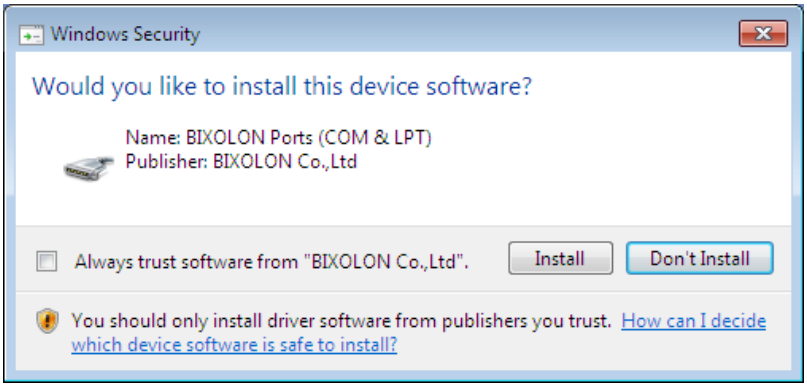
2) Click the “Next” button.



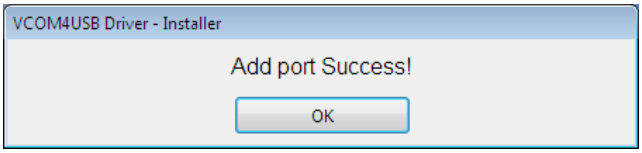
3) Clicking the “Install” button begins the installation process.



4) When the following message appears, click the “Install” button.



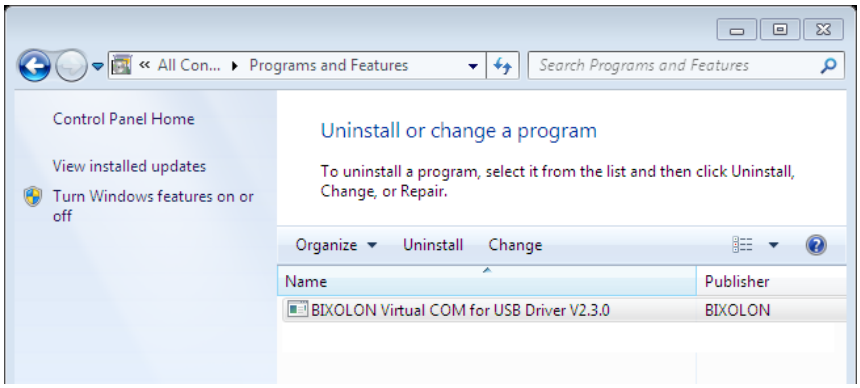
5) The following message appears upon the successful completion of installation.



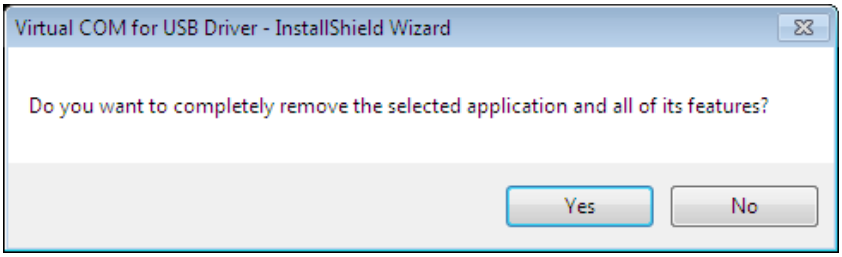
3-2 Uninstallation

1) Click and activate the “Uninstall a program” or “Programs and Features” icon in the Control Panel.

2) Select “Virtual COM for USB Driver” and click the “Uninstall“ button.



3) Click the “Yes” button to completely remove the application.

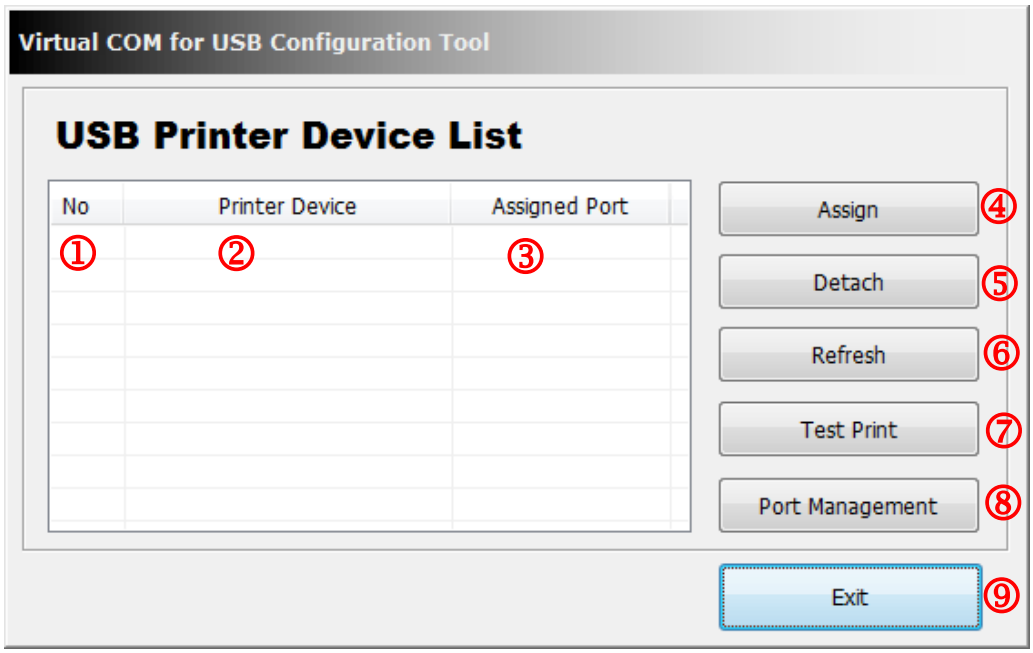


4. Configuration Tool Usage

The “Configuration Tool” program can be used to create a serial port, and set and/or delete printer devices.

4-1 Configuration Tool Functions

Select from “Start ▶ All Programs ▶ BIXOLON ▶ Virtual COM for USB Driver” and open “Configuration Tool”.



The “Configuration Tool” functions are described as follows.

① Printer Device Sequence Number
- This function displays the connection sequence of printer devices.

② Printer Device Name
- This function displays the printer device models that are turned on and connected via the USB interface.

③ Assigned Port
- This function displays the currently set virtual serial port number. If one is not set, the status is shown as “Not Assigned”

④ Assign
- This function is used to establish a connection between a printer selected from the list box and a registered virtual serial port.

⑤ Detach
- This function is used to disconnect a connection between a printer selected from the list box and a virtual serial port.

⑥ Refresh

- This function is used to update the USB printer device list shown in the list box and set port information.

⑦ Test Print

- This function is used to perform a print test on the printer selected from the list box. The serial port must be set in order to perform a print test.

⑧ Port Management

- This function is used to add/remove virtual serial ports.

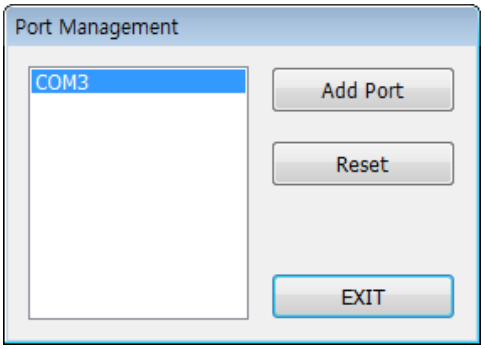
⑨ Exit

- This function is used to exit the “Configuration Tool” program.

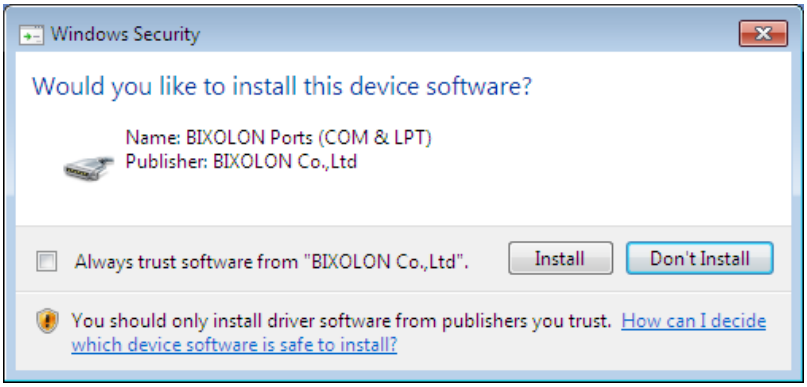
4-2 Virtual COM Port Addition

The “Configuration Tool” program is used to add a virtual COM port.
To create a port, click the “Port Management’ button in the “Configuration Tool”.

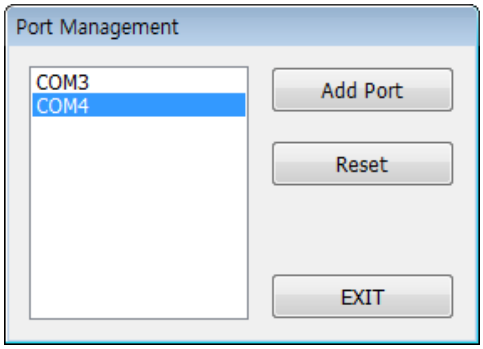
1) Clicking the “Port Management” button of the “Configuration Tool” brings up the following dialog box.



2) Click the “Add Port” button to add a port.
Click “Install” when “Windows Security” screen below appears.



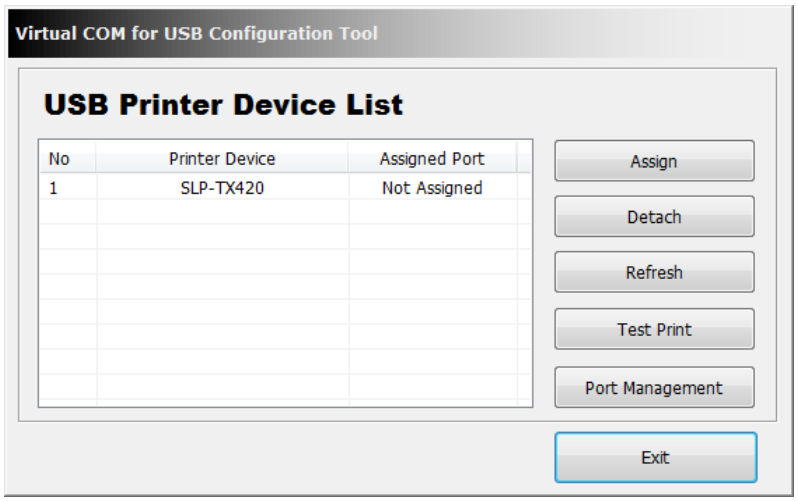
3) If the port has been added successfully, the virtual serial port added will appear as follows.



※ NOTE
- When adding a virtual serial port via the “Add Port” function, the port number indicated in “COM#” is determined according to the operating system. Refer to section 5 (Virtual Serial Port Management) of this manual for information on modifying the virtual serial port number.

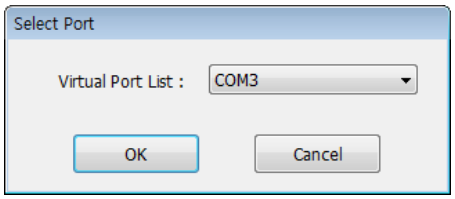
4-3 Printer Port Connection

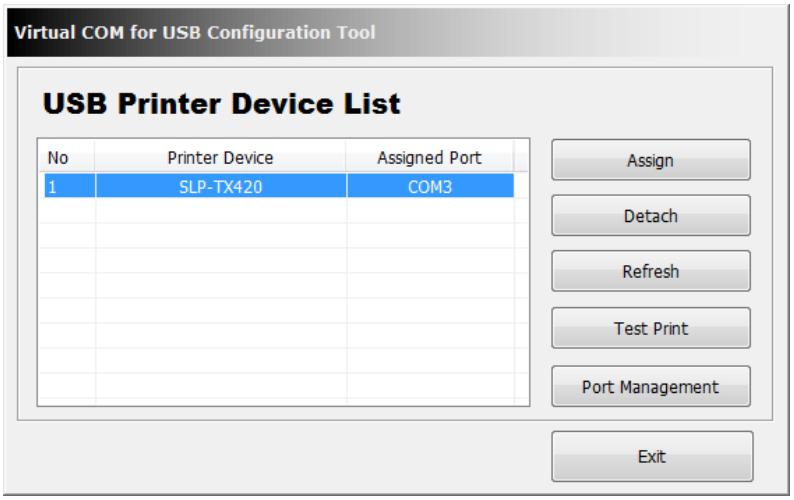
To use the added serial port and printer, click the “Assign” button and set the connection.



1) Select the printer to be connected from the list, and click the “Assign” button to bring up a dialog box.

2) Select the port number to be used among the virtual serial ports added via “Port Management” and click the “OK” button to automatically update the “Assigned Port” section of the main screen.



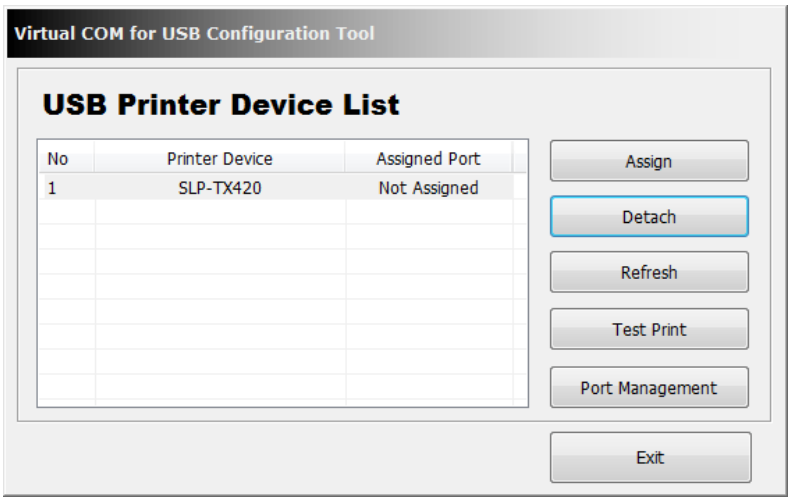


※ NOTE

- When connecting a virtual serial port, if another printer is connected to the corresponding port, that existing connection is automatically disconnected, and the connection is established with the selected printer.

4-4 Printer Port Disconnection

Select the printer to be disconnected from the list of printers connected via virtual serial ports. Click the “Detach” button to disconnect the previously connected virtual serial port. This will result in reversion of the “Assigned Port” status of the printer to “Not Assigned”.



※ NOTE

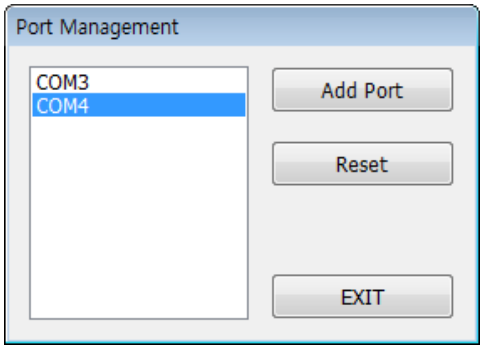
- Although the port setting of the printer has been disconnected, the virtual serial port is not deleted.
- The “Port Management” function can be used to delete all ports, or the Device Manager can be used to manually delete select ports.

4-5 Print Test

The “Test Print” button can be used to check the communication status between the printer and the virtual serial port.

4-6 Virtual Serial Port Deletion

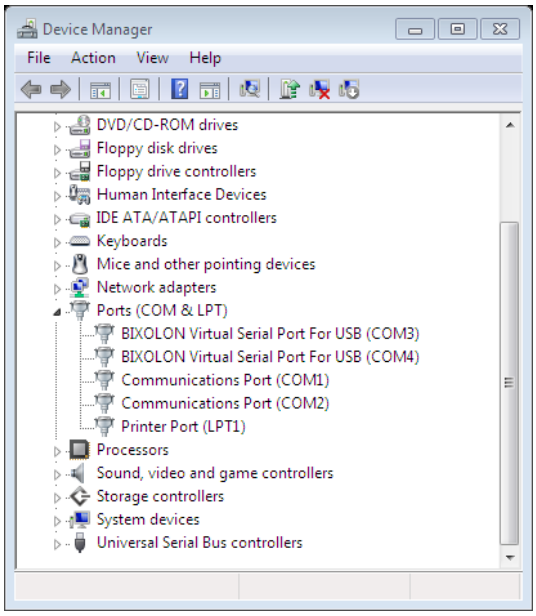
Click the “Port Management” button to delete the connection settings of virtual serial ports. The “Reset” button can be used to delete all virtual serial ports and connected printer information.



5. Virtual Serial Port Management

Port number modification, port deletion, and serial communication setting are done via the Device Manager.

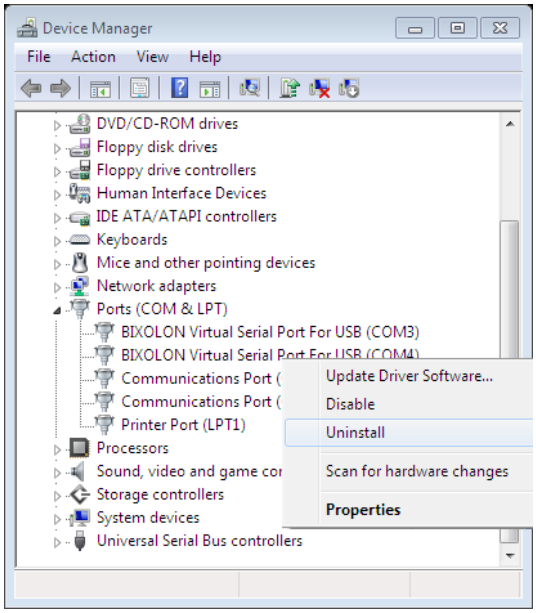
Control Panel ▶ System ▶ Device Manager: Upon examination of the ports (COM & LPT) section, the port(s) added previously via the “Configuration Tool” program can be seen.



5-1 Port Deletion Via the Device Manager

Select the virtual serial port to be deleted and click the right mouse button. On the popup menu, select “Uninstall” to delete the port.

Following deletion, the “Configuration Tool” program must be executed, and the changes must be updated.

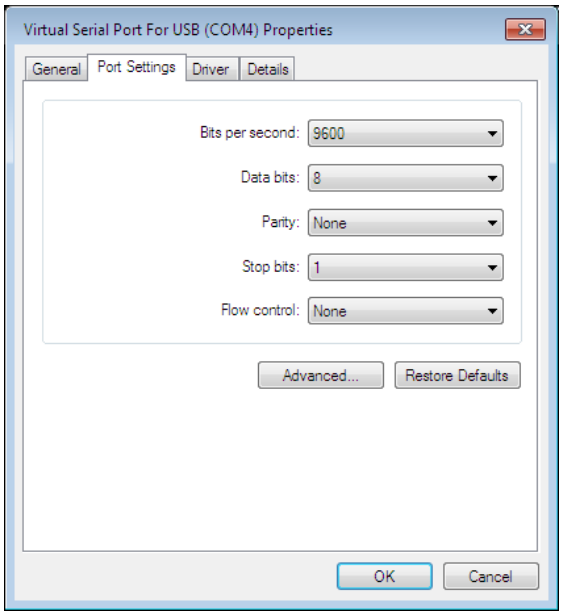


5-2 Port Setting/Number Modification Via the Device Manager

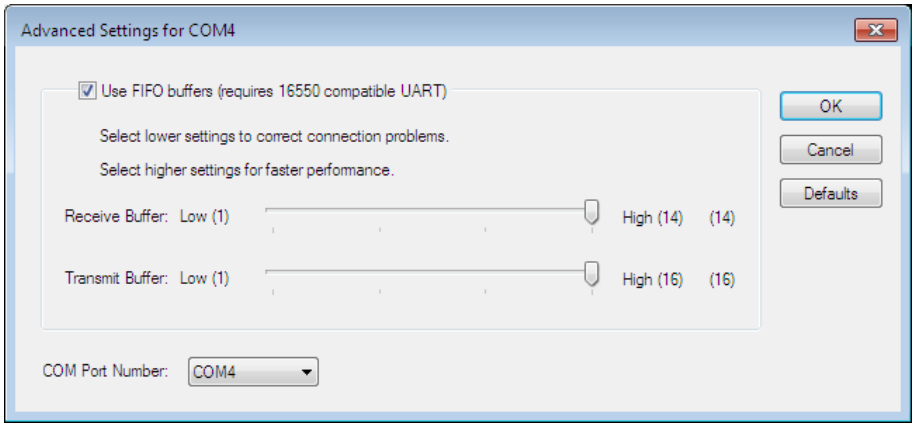
1) Select the desired virtual serial port and click the right mouse button. On the popup menu, select “Properties”.

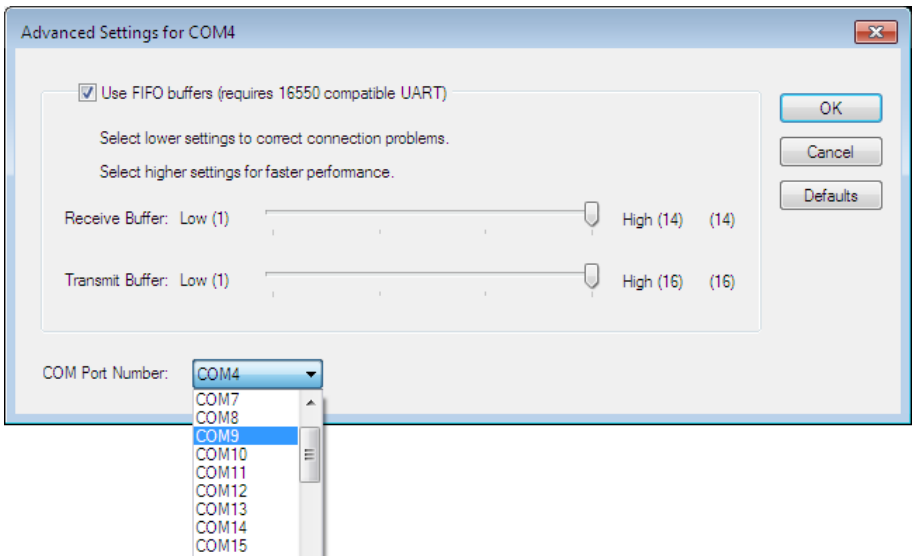
2) In the “Properties” window, click the “Port Settings” tab.

3) The “Port Settings” tab can be used to set the baud rate, data bits, parity, stop bits, and flow control of a port.



4) Click the “Advanced” button to modify the port number and/or designate buffer settings. Modification of buffer settings is not recommended. The COM port number is changed by selecting a number in the combo box.





Following port modifications, the settings utility must be run to update changes.
If a printer was connected previously, the connection setting must be disconnected.

※ CAUTION
The selecting port must not already be in use.

Following port modification, clicking the “Yes” button can result in the appearance of a warning message. Click the “Yes” button to finalize and set the modified information and settings. Port modification on certain operating systems will require the repeat opening of the Device Manager to confirm changes.

After deleting a virtual serial port, the “Configuration Tool” program must be executed to update the Virtual COM for USB Driver, and directly deleted connection information must be removed.

6. Win32 API

The “Virtual COM for USB Driver” program can be used in serial communication programming. Refer to Microsoft MSDN regarding the Win32 API for serial communication.

※ NOTE
Currently, only basic serial communication APIs are supported. In the future, additional APIs suited to user needs are set to be added.

6-1 File I/O API

1) CreateFile(): Supported
2) WriteFile():Supported
3) ReadFile():Supported
4) CloseHandle():Supported

6-2 Communication API

1) BuildCommDCB():Supported
2) BuildCommDCBAndTimeouts():Supported
3) CommConfigDialog():Supported
4) DeviceIoControl():Supported
5) GetCommModemStatus():Supported
6) GetCommState():Supported
7) GetCommTimeouts():Supported
8) PurgeComm():Supported
9) SetCommState():Supported
10) SetCommTimeouts():Supported

6-3 Precautions for Serial Communication Programming

1) With CreateFile, any port name ranging from 1 to 9 may be used as “COM1” to “COM9”. To use ports 10 to 256, the following format must be used, “\\\\.\\COM10” to “\\\\.\\COM256”.